

Case Study No.: 2

Case Study Title

Online Shopping Billing System using HTML, CSS, HTML5, and JavaScript

Theory

An **Online Shopping Billing System** is a web-based application that automates the process of generating customer bills for purchased products. It collects customer details, product information, pricing, discounts, taxes, and payment details, then calculates the final payable amount accurately. Such systems reduce manual calculation errors, save time, and improve billing efficiency in retail stores, supermarkets, and e-commerce platforms.

This case study is developed using **HTML5**, **CSS3**, and **JavaScript**.

HTML5

HTML5 is used to create the structure of the webpage. It provides form elements such as text boxes, number fields, and drop-down lists to collect customer information including customer name, mobile number, invoice number, product name, quantity, rate per kilogram, discount, GST, packing charges, payment mode, and membership status.

CSS3

CSS3 is used to design the user interface of the application. It improves the appearance of the webpage by applying colors, fonts, spacing, borders, buttons, and responsive layouts, making the billing system more attractive and user-friendly.

JavaScript

JavaScript is used to add functionality to the webpage. It accepts user inputs, performs calculations, applies business logic, and dynamically generates the shopping invoice without refreshing the page.

Billing Calculation

The billing system performs the following calculations:

- **Total Amount**

$$\text{Total} = \text{Quantity} \times \text{Rate per Kg}$$

- **Discount Amount**

$$\text{Discount Amount} = (\text{Total} \times \text{Discount \%}) \div 100$$

- **Amount after Discount**

$$\text{Amount after Discount} = \text{Total} - \text{Discount Amount}$$

- **GST Amount**

$$\text{GST Amount} = (\text{Amount after Discount} \times \text{GST \%}) \div 100$$

- **Final Bill**

$$\text{Final Bill} = \text{Amount after Discount} + \text{GST Amount} + \text{Packing Charges}$$

- If the customer has a **Membership**, an additional discount (for example ₹100) is deducted from the final bill.

Features of the System

- Accepts customer and product details.
- Calculates the total purchase amount.
- Applies discount and GST automatically.
- Adds packing charges to the bill.
- Supports multiple payment methods such as **Cash**, **Card**, and **UPI**.
- Provides membership benefits by applying an additional discount.
- Generates a detailed invoice dynamically on the webpage.
- Simple, accurate, and easy to use.

Applications

- Grocery Stores
- Supermarkets
- Retail Shops
- E-commerce Websites
- Departmental Stores

- Wholesale Shops

Advantages

- Reduces manual calculation errors.
- Generates bills quickly and accurately.
- Improves customer service.
- Saves time and effort.
- Provides a professional billing system.
- Easy to modify and maintain.
- Enhances user experience through an interactive web interface.

1. Case Study Program Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width,initial-scale=1">

<title>Online Shopping Billing System</title>

<style>

body{
font-family:Arial;
background:#eef5ff;
padding:40px;
}

.container{
max-width:700px;
margin:auto;
background:white;
padding:25px;
border-radius:10px;
box-shadow:0 0 15px gray;
```

```
}

h2{
text-align:center;
color:#0b5394;
}

label{
font-weight:bold;
}

input,select{
width:100%;
padding:10px;
margin-top:5px;
margin-bottom:15px;
}

button{
width:100%;
padding:12px;
background:#0b5394;
color:white;
border:none;
font-size:18px;
cursor:pointer;
}

button:hover{
background:#073763;
}

#bill{
margin-top:25px;
background:#f4f4f4;
padding:20px;
border-radius:8px;
}

</style>
```

```
</head>

<body>

<div class="container">

<h2>Online Shopping Billing System</h2>

<label>Customer Name</label>
<input type="text" id="name">

<label>Mobile Number</label>
<input type="tel" id="mobile">

<label>Invoice Number</label>
<input type="text" id="invoice">

<label>Product Name</label>
<input type="text" id="product">

<label>Quantity (Kg)</label>
<input type="number" id="qty">

<label>Rate per Kg (₹)</label>
<input type="number" id="rate">

<label>Discount (%)</label>
<input type="number" id="discount">

<label>GST (%)</label>
<input type="number" id="gst">

<label>Packing Charges (₹)</label>
<input type="number" id="packing">

<label>Payment Mode</label>

<select id="payment">
<option>Cash</option>
```

```
<option>Card</option>
<option>UPI</option>
</select>

<label>Membership</label>

<select id="member">
<option>No</option>
<option>Yes</option>
</select>

<button onclick="calculateBill()">Generate Bill</button>

<div id="bill"></div>

</div>

<script>

function calculateBill(){

let name=document.getElementById("name").value;
let mobile=document.getElementById("mobile").value;
let invoice=document.getElementById("invoice").value;
let product=document.getElementById("product").value;

let qty=Number(document.getElementById("qty").value);
let rate=Number(document.getElementById("rate").value);
let discount=Number(document.getElementById("discount").value);
let gst=Number(document.getElementById("gst").value);
let packing=Number(document.getElementById("packing").value);

let payment=document.getElementById("payment").value;
let member=document.getElementById("member").value;

let total=qty*rate;

let discountAmount=(total*discount)/100;

let subtotal=total-discountAmount;
```

```
let gstAmount=(subtotal*gst)/100;

let finalBill=subtotal+gstAmount+packing;

if (member=="Yes") {

finalBill=finalBill-100;

}

document.getElementById("bill").innerHTML=`

<h3>Invoice</h3>

<b>Customer :</b> ${name}<br>
<b>Mobile :</b> ${mobile}<br>
<b>Invoice :</b> ${invoice}<br>
<b>Product :</b> ${product}<br>
<b>Quantity :</b> ${qty} Kg<br>
<b>Rate :</b> ₹${rate}<br>
<b>Total :</b> ₹${total.toFixed(2)}<br>
<b>Discount :</b> ₹${discountAmount.toFixed(2)}<br>
<b>GST :</b> ₹${gstAmount.toFixed(2)}<br>
<b>Packing :</b> ₹${packing}<br>
<b>Payment :</b> ${payment}<br>
<b>Membership :</b> ${member}<br>

<hr>

<h2>Final Bill : ₹${finalBill.toFixed(2)}</h2>

`;

}

</script>

</body>
</html>
```


Output:

Online Shopping Billing System

Customer Name

RADHIKA SURESHRAO CHAUDHARI

Mobile Number

9405606890

Invoice Number

32

Product Name

Mac

Quantity (Kg)

1

Rate per Kg (₹)

1000000

Discount (%)

20

GST (%)

20

Packing Charges (₹)

5000

Payment Mode

UPI



Membership

No



Generate Bill



Result/Conclusion:

The experiment and case study were successfully implemented using **HTML5, CSS3, and JavaScript**. Primitive data types such as **String, Number, Boolean, Undefined, and Null** were demonstrated along with type conversion. An **Online Shopping Billing System** was developed that accepts customer and product details, calculates the total amount after applying discount, GST, packing charges, and membership benefits, and generates a detailed invoice. This case study improved the understanding of HTML form elements, JavaScript data types, arithmetic operations, user input handling, and dynamic webpage generation.